CHAPTER 12

Parametric representations and Cartesian equations

S65, S67, S70

Before you begin · Lines, planes and systems

Without a calculator, in six minutes. Answer briefly. Use direction vectors, normal vectors and membership. Then choose the targeted follow-up indicated under the questions.

1. Give a direction vector of (AB) for A=(1;2;3), B=(3;1;3).

2. Check that (1;2;-1) is orthogonal to (2;-1;0) and (1;0;1).

3. Is the vector (1;1;1) orthogonal to (1;2;-1)?

4. Solve 2 $t = 6$.

5. What must be checked for a point to be the projection of M onto a plane?

6. Are the vectors (1;0;0) and (0;1;0) collinear?

Answers

$(2; -1; 0)$; both products equal 0; no, the product equals 2; $t = 3$; membership of the plane and orthogonality; no, no multiple of the first gives the second.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Subtract A from B, coord. by coord.; R05. **2.** Calculate the two scalar products separately; R60. **3.** Add the three products and compare to zero; R60. **4.** Divide both sides by 2 and check. **5.** Go back over the two distinct conditions for the projection; R63. **6.** Compare the first then the second coordinates; R06. The R questions serve as review prompts with the teacher, even if their collective routine is planned for later.

Objectives

Describe a line and a plane; find intersections; choose a method for a projection.

Course notes

Throughout this chapter, coordinates are given in an orthonormal coordinate system of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \quad (t \in \mathbb{R}).$$

A point lies on the line if there exists one real t satisfying all three equalities. A line has many representations: changing the starting point or multiplying the direction vector by a non-zero real number.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Plane equations

Equation of a plane

If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a, b, c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, so it belongs to the plane.

Example. The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is 2 $(x - 1) - (y - 2) + 3z = 0$, i.e. $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single point of intersection; an identity gives a line contained in the plane; a contradiction gives a disjoint parallel line. For two planes, non-collinear normal vectors give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normal vectors are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use parameters s, t (independent) and solve the three equations. With no solution and non-collinear direction vectors, the lines are non-coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, find the projection of M on the normal line through M : $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learnt in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Checking is essential. After calculating a projection, check simultaneously $H \in \mathcal{P}$ and the collinearity of $\overrightarrow{MH}$ with the normal vector.

Moving from objects to equations

12.1 One line, several parametrisations

First use: S65.

Definition · Parametrising a line

A line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$, with $t \in \mathbb{R}$. The parameter is neither a prescribed coordinate nor a time, unless the context specifies it.

Worked example · Same set, other parameters

The expressions $(1+t;2-t;2t)$ and $(3+2s;-2s;4+4s)$ describe the same line: set $t = 2+2s$ transforms the first into the second, and this transformation runs through all real numbers. The starting points may differ, and the direction vectors may be proportional.

A1 · Immediate application Show that $(t;1+2t;3-t)$ and $(2+3s;5+6s;1-3s)$ describe the same line. Is the point $(1;3;2)$ belong to it?

12.2 A normal vector fixes the equation of a plane

First use: S67.

Property · Cartesian equation

In an orthonormal coordinate system, a plane with nonzero normal vector $(a;b;c)$ through $A = (x_A;y_A;z_A)$ has equation $a(x-x_A)+b(y-y_A)+c(z-z_A)=0$. It expresses $\overrightarrow{AM}\cdot\vec{n}=0$.

Worked example · Construct, then test

For A=(1;0;2) and $\vec{n}=(2;-1;3)$, the plane has equation 2 $x-y+3z-8=0$. The point (0;1;3) lies in the plane since 0 $-1+9-8=0$. A point does not lie in the plane just because its triple resembles the normal vector.

A2 · Partial guidance Determine the plane with normal vector $(1;2;-1)$ through (2;0;1). Test whether (1;1;2) lies in it, then give two non-collinear directions of the plane.

12.3 Three outcomes for a single equation

First use: S69.

After substitution	Interpretation	Example
$at+b=0,\ a\neq 0$	a point	intersecting line
$0t+b=0,\ b\neq 0$	no point	strictly parallel
$0t+0=0$	all points	line in the plane

Worked example · Reading the equation y = mx + p

For the plane x+y+z=1 and the line $(t;2t;1-t)$, substitution gives 2 $t+1=1$, so t=0 and a single intersection (0;0;1). We conclude only after substituting the value back into all three coordinates.

A3 · Compare In the plane $z=1$, study the relative positions of the lines $d_1 : (t;0;t)$, $d_2 : (t;0;0)$ and $d_3 : (t;0;1)$. Give the intersection point when it is unique and sketch the three situations.

12.4 Two lines, two independent parameters

First use: S70.

Worked example · Do not impose a false simultaneity

For $d : (t;0;0)$ and $d' : (1;1+s;0)$, the intersection requires t=1 and s=-1. Imposing t=s would wrongly eliminate the point (1;0;0). The parameters are used separately to run along each line.

A4 · Fix a method For $d : (t;t;0)$ and $d' : (1+s;1-s;s)$, solve with two parameters, then explain why all three coordinates must be checked.

12.5 Two checks after the system

First use: S70.

Worked example · Projecting onto a plane

Let $\mathcal{P} : x+2y-z=1$ and M=(1;1;1). The normal line through M is $H=(1+t;1+2t;1-t)$. Substituting into the plane gives 2 $+6t=1$, therefore $t=-1/6$. H=(5/6;2/3;7/6). We check 5 $/6+4/3-7/6=1$ and $\overrightarrow{MH}=(-1/6)(1;2;-1)$. The distance is $\overline{6}/6$.

A5 · Synthesis with a degenerate case For a real number m, d_m passes through (0;0;m) and has direction $(1;m;1)$. Study the intersection with $x+2y-z=1$. Treat m=0 separately. Is there an m for which the line is perpendicular to the plane?

A first step: Substitute the three coordinates before dividing. For orthogonality, compare the direction vector with a normal vector.

A6 · Choice of strategy Determine the intersection of the planes $x+y+z=3$ and $x-y=0$. Give a point and a direction vector, then check these two pieces of information in the equations.

One minute, spoken aloud. State the two properties that make it possible to recognise a projection without mentioning coordinates.

Delayed review · S71 then S87. Match each line of an algebraic calculation with a geometric property; deal first with the parameters that make a denominator zero.

Investigation

Three lines and a plane - 30 min.

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x+y+z=3$. In each question, the parameter t runs through $\mathbb{R}$.

1. The line d_1 has the representation $(x;y;z)=(t;t;t)$. Determine $d_1 \cap \mathcal{P}$.

2. The line d_2 has the representation $(x;y;z) = (3+t;-t;0)$. Determine $d_2 \cap \mathcal{P}$.

3. The line d_3 has the representation $(x;y;z) = (t;-t;0)$. Determine $d_3 \cap \mathcal{P}$.

4. For each line, state whether the equation obtained after substituting into the plane's equation has a unique solution, no solution or every real number as a solution. Relate these cases to the geometric positions found.

Progressive exercises

E1 · Direct application. In a coordinate system of space, consider $A(1;-1;2)$ and $B(3;0;1)$. Give a parametric representation of the line (AB).

E2 · Application and justification. In an orthonormal coordinate system, determine a Cartesian equation of the plane passing through $A(1;2;3)$ with normal vector $\vec{n} = (1;0;-2)$.

E3 · Method to organise. In a coordinate system of space, the line d is given by $(x;y;z) = (t;1+t;2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x+2y+z=8$. Determine their intersection.

E4 · Choice of strategy. In a coordinate system of space, determine the intersection of the planes with equations $x+y+z=1$ and $x-y=0$. Give a parametric representation if the intersection is a line.

E5 · Synthesis or proof. In a coordinate system of space, study the relative position of the planes with equations 2 $x-y+z=3$ and $4x-2y+2z=5$. Justify.

Work from the textbook or a past paper

Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5-6). At home after S70. Justify each answer; for a false statement, explain precisely what disproves it. The paper is available here: official past paper . Use the data and instructions of this past paper.

Homework

DM12 - Projection and intersection - 55 min.

In an orthonormal coordinate system of space, consider the point $M(3;0;0)$ and the planes $\mathcal{P} : x+2y+2z = 9$ and $\mathcal{Q} : x-y=0$.

1. Give the coordinates of a nonzero normal vector to $\mathcal{P}$.

2. Determine the coordinates of the orthogonal projection H of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.

3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.

4. The point H lie on this line? Justify.

5. Determine the coordinates of the point K of this line whose z-coordinate is zero. Calculate MK, then compare MK and MH.

MCQ A · reading and checking a representation

Orthonormal coordinate system. After S67; MCQ B after S70.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $d : (x;y;z) = (1;0;2) + t(2;-1;0)$ has direction vector:

A. $(1;0;2)$ B. $(2;-1;0)$

C. $(2;1;0)$ D. $(0;0;2)$

2. Which of these points belongs to d ?

A. $(3;1;2)$ B. $(1;0;0)$

C. $(2;-1;0)$ D. $(3;-1;2)$

3. A normal vector to the plane $x + 2y - 2z = 3$ is:

A. $(1;2;-2)$ B. $(1;2;3)$

C. $(3;0;0)$ D. $(-2;1;0)$

4. To check that H is proj. of M on $\mathcal{P}$, we need:

A. only $H \in \mathcal{P}$ B. only MH small

C. $H \in \mathcal{P}$ and $\overrightarrow{MH}$ normal to the plane D. that $H = M$

5. The line $A + tu$ with $u \neq 0$ is also represented by:

A. $A + t\vec{0}$ B. $A + 2u + s(3u)$

C. $tA + u$ D. the single point $A + u$

MCQ B · intersections and special cases

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Substituting a line into a plane equation gives $0 = 4$. The intersection is:

A. a line B. a point

C. the plane D. empty

7. If substitution gives $0 = 0$ for every parameter value:

A. the line lies in the plane B. there is no common point

C. there is exactly one common point D. the data are wrong

8. To find the intersection of $A + tu$ and $B + sv$, we must:

A. set $s = t$ B. replace both parameters with 0

C. solve for s, t independent D. ignore the third coordinate

9. The intersection of $x + y = 1$ and $z = 2$ is:

A. a plane B. the line $(t;1-t;2)$

C. the point $(1;0;2)$ D. empty

10. For $m = 0$, the set $mx + y = 1$ is:

A. empty B. a line

C. the whole of space D. the plane $y = 1$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Two lines whose direction vectors are not collinear necessarily intersect.

2. The planes $x + y + z = 1$ and $2x + 2y + 2z = 3$ coincide.

3. The projection of M onto a plane is unique.

MCQ solutions

1. **B.** The vector multiplying t is a direction vector; the first triple denotes a point on the line.
2. **D.** Take $t = 1$ satisfies all three coordinates simultaneously. A single correct coordinate does not suffice.
3. **A.** In an orthonormal coordinate system, the coefficients of x, y, z give a normal vector. The constant term is not its third coordinate.
4. **C.** These two conditions characterise the projection. If M is in the plane, the zero vector works and $H = M$.
5. **B.** The new point is on the line, and 3 u is a non-zero direction vector collinear with u. The change $t = 2 + 3s$ runs through all $\mathbb{R}$.
6. **D.** No value of the parameter works: the line is strictly parallel to the plane.
7. **A.** Every point of the line satisfies the equation of the plane. The intersection is the whole line.
8. **C.** The same point can be reached for two different values of the parameters. Three coordinates must be checked.
9. **B.** The two constraints leave one free parameter; taking $x = t$ forces $y = 1 - t$ and $z = 2$.
10. **D.** The coefficient of y remains non-zero. An equation can still define a plane when a coefficient is zero.

True/false. False: they may be non-coplanar. False: they have collinear normal vectors but incompatible constants; the planes are strictly parallel. True: the normal line through M intersects the plane at a single point.

Solutions to the applications and guidance

Application 1. Let t=2+3s; substitution gives the three coordinates of the second parametrisation and runs through $\mathbb{R}$. For (1;3;2), t=1 satisfies the three coordinates.

Application 2. Equation $x+2y-z=1$. The proposed point gives $1 \quad +2-2=1$, so it belongs to the plane. Possible directions $(2;-1;0)$ and $(1;0;1)$, orthogonal to the normal and not collinear.

Application 3. d_1 intersects at (1;0;1); d_2 is strictly parallel; d_3 lies in the plane. Substitution gives t=1, 0=1 and 1=1, respectively.

Application 4. The equations are t=1+s, t=1-s and 0=s. So s=0, t=1, point (1;1;0). Solving two equations can produce candidates; only checking the third one is conclusive.

Application 5. $d_m : (t; mt; m+t)$. The equation becomes $2mt-m=1$. If m=0, it is impossible: the line strictly parallel. Otherwise $t=(m+1)/(2m)$ provides a single point. To be perpendicular, $(1;m;1)$ would have to be collinear with (1;2;-1); the first coordinate forces a ratio of 1, incompatible with the third. No m.

Application 6. x=y=t and z=3-2t, so line $(t;t;3-2t)$, point (0;0;3), direction (1;1;-2). The point satisfies both equations; the direction vector is orthogonal to both normals (1;1;1) and (1;-1;0).

From the system to geometry

Have students predict the three cases from the figure, then read the equations 0=1 and 0=0 as useful results. To differentiate, provide the substitution to a student who has grasped the idea but gets lost in the calculation.

Teaching notes and solutions

Guidance for the teacher

Use two different parameters to compare two lines. Check the three coordinates, even though two are enough to compute candidates. A Cartesian equation alone generally describes a plane in space; it does not describe a line. Reuse the projection from chapter 11 to give meaning to solving systems.

Solutions to the activity

1. A point of d_1 has coordinates (t,t,t). It lies in the plane if and only if 3 $t=3$, i.e. $t=1$. The intersection is therefore the unique point $(1,1,1)$.

2. For a point of d_2, $x+y+z=3+t-t+0=3$ whatever t. All points of the line are in the plane: $d_2 \subset \mathcal{P}$.

3. For a point of d_3, $x+y+z=t-t+0=0$. The equation $0=3$ has no solution: the line is disjoint from the plane. Its direction vector $(1,-1,0)$ is orthogonal to the normal vector $(1,1,1)$; it is parallel to the plane.

4. After substitution, we obtain respectively an equation with a single solution, an identity true for every value of the parameter, and a contradiction. These three cases correspond to an intersection at a point, inclusion and no intersection.

E1. $\overrightarrow{AB}=(2,1,-1)\neq\vec{0}$. The line therefore has the representation $(x,y,z)=(1,-1,2)+t(2,1,-1)$, $t\in\mathbb{R}$. The values $t=0$ and $t=1$ give back A and B.

E2. For $A(1,2,3)$, the condition $\overrightarrow{AM}\cdot(1,0,-2)=0$ is $(x-1)-2(z-3)=0$. After expanding, the plane has equation $x-2z+5=0$. The point A satisfies it.

E3. After substitution, $t+2(1+t)+(2-t)=8$, i.e. 2 $t+4=8$, so $t=2$. The unique intersection is $(2,3,0)$; check 2 $+2\times3+0=8$.

E4. The second equation gives $x=y$. Set $x=y=t$; the first becomes 2 $t+z=1$. The set of solutions is $(x,y,z)=(t,t,1-2t)=(0,0,1)+t(1,1,-2)$, a line.

E5. The normal vectors are collinear, the second being twice the first. But doubling 2 $x-y+z=3$ gives 4 $x-2y+2z=6$, incompatible with the right-hand side 5. The planes are therefore strictly parallel, and not intersecting.

Solutions to homework DM12

1. The coefficients of x, y, z give the non-zero normal vector $\vec{n} = (1, 2, 2)$ in the orthonormal coordinate system.

2. Let us look for H on the normal line through $M : H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We check $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. Hence H is indeed the projection, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.

3. The normal vectors $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the planes intersect in a line. Since $x = y$, let $x = y = s$. The first equation gives $3 s + 2z = 9$, i.e. $z = \frac{9-3s}{2}$. The intersection is the line $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.

4. The point H : $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it does not belong to $\mathcal{Q}$ nor to the line of intersection.

5. The z-coordinate is the coordinate z. Setting it to zero requires $9 - 3s = 0$, therefore $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result agrees with the property of the projection: among all points of the plane, H is closest to M.

Solutions to the textbook or past-paper work

1. True: the line passes through B and its direction vector $(-2; -1; 3)$ is $-\frac{1}{2}\overrightarrow{AB}$.

2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.

3. False: for $k = -4$ and $s = \frac{5}{2}$, both representations give $(11; 12; -14)$. The lines intersect, so they are coplanar.

4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.